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AREA: ELECTRICAL	DESCRIPTION : ELECTRICAL DESIGN BASIS	

IP-009

PASHMINA PROJECT

ELECTRICAL DESIGN BASIS

MOMENTIVE PERFORMANCE MATERIALS

CHENNAI

REFERENCE SPEC. NO. CCD98-43-00-01 REV. P					
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1.0 SCOPE

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The intention of this document is to describe the project requirements and principles for the electrical design and equipment selection for Pashmina project, Oragadam, Sriperambudur taluk, Kancheepuram district, Tamilnadu Approx. 50kms from Chennai.

This document does not provide the scope of design, documentation and equipment supply.

The design shall be based on following factors :

- Safety of personnel and equipment during operating and maintenance.
- Reduce fire and explosion hazards due to short circuits, faults, process mishaps and defective equipment.
- Reliable, secure, efficient & Convenience of operation.
- Ease of maintenance.
- Minimum fire risk.
- Maximum interchangeably of equipment.
- Latest technology.
- Safe Industrial Practices and convenience of operation.

2.0 GENERAL**2.1 STANDARDS AND CODES**

All electrical equipment shall be designed, manufactured, tested, installed and commissioned in compliance with the latest revisions of following codes & standards including addenda , revisions or supplements.

- | | |
|-------|---|
| [] | ANSI, IEEE, NEMA, NEC, NESC, UL |
| [X] | IEC |
| [X] | BUREAU OF INDIAN STANDARDS |
| [X] | OTHERS: LOCAL STATUTORY REGULATIONS, INDIAN ELECTRICITY RULES – 1956, INDIAN ELECTRICITY ACT-1910 |

2.2 SITE CONDITIONS

- | | | | |
|-------|-------------------------------|---|---------------------------|
| 2.2.1 | Equipment Design temp | : | 45 Deg. C. |
| 2.2.2 | Mean Relative humidity | : | 88% max, 77% min. |
| 2.2.3 | Monthly heaviest Rainfall | : | 143 mm |
| 2.2.4 | Soil Resistivity in Ohm – mtr | : | (Will be furnished later) |
| 2.2.5 | Altitude above mean sea level | : | 16 Mtrs. |
| 2.2.6 | Monthly mean wind speed | : | 10 KMPH |
| 2.2.7 | Sesmic zone | : | Zone III |

2.3 POWER SUPPLY CONDITIONS**2.3.1 GRID SUPPLY**

- | | | | |
|----|-------------------------|---|----------------------------|
| a) | Name of grid substation | : | 220/110/33/11KV – Oragadam |
| b) | Number of feeders | : | one |

c)	Capacity of 11KV feeder	:	Present + Technology center load (Future expansion is not considered)	REV.
d)	Minimum load PF stipulated By supply authorities	:	0.92	
e)	Plant Supply Voltage	:	11KV AC supply.	
f)	Variation of voltage supply	:	±10 %	
g)	Frequency	:	50 Hz	
h)	Variation of Frequency	:	± 3%	
i)	Expected Grid Fault level at 11KV	:	250MVA (Assumed value to confirmed later)	

2.3..2 ALTERNATE POWER SUPPLY

a)	Alternate power source	:	Diesel Generator sets
b)	Capacity of DG sets	:	To cater power supply to Present load (Excluding Technology center)
c)	Provision for future expansion:	:	Required
d)	DG supply Voltage	:	415V
e)	No. of DG sets envisaged	:	One "OR" Two
f)	Export of power to Grid	:	Not envisaged

2.4 UTILIZATION VOLTAGES

Levels of utilization voltages in the plant are as follows.

2.4.1	Voltage of plant Low Voltage Distribution Network (LVDN)	:	415V, 50Hz, 3ph, 4W Neutral solidly Earthed (Transformer Earthed Neutral)
2.4.2	Motors < 0.37KW	:	230V, 50 Hz, 1ph, 3W
	Motors upto & including 200 KW	:	415V; 50 Hz; 3ph; 3W
	Motor above 200 KW	:	415V; 50 Hz; 3ph; 3W
2.4.3	Heaters < 3 KW	:	230 V, 50 Hz, 1 phase
2.4.4	Heaters > 3 KW	:	415 V; 50Hz, 3 phase
2.4.5	Auxiliary supply for Switchyard equipments / MV Indoor Switchboard		
	MV Indoor & Outdoor breaker		
	a) Closing, tripping, indication	:	110 V DC Supply
	MV Outdoor Disconnecting switch		
	a) Opening, Closing & interlocking	:	110 V DC Supply
2.4.6	Motor starter Control Voltage	:	110 V ± 10% ; 50 Hz ; 1ph ; 2W Grounded Neutral. (Derived from each switchboard)
2.4.7	LV breaker		

a)	Closing, spring charging.	:	110 V $\pm$ 10% ; 50 Hz ; 1ph ; 2W Grounded Neutral.	REV.
b)	Tripping, Indication & Protection relay aux supply	:	110 V DC Supply	
2.4.8	Lighting Fixtures	:	230 V, 50 Hz, 1ph, 3W, AC Supply	
2.4.9	Lighting Receptacles	:	230 V, 50 Hz, 1ph, 3W, AC Supply	
2.4.10	Welding Receptacles	:	415 V, 50 Hz, 3ph, 5W, AC Supply	
2.4.11	DCS Power Supply	:	110V; 50 Hz; 1ph; 3W, AC Supply Grounded Neutral (From UPS system)	
2.4.12	Instrumentation Power Supply	:	110V; 50 Hz; 1ph; 3W, AC Supply (From UPS system)	
2.4.13	Fire alarm system power supply	:	110V; 50 Hz; 1ph; 3W, AC Supply (From UPS system)	
2.4.14	Telephone system power supply	:	110V; 50 Hz; 1ph; 3W, AC Supply (From UPS system)	
2.4.15	Public address system power supply	:	110V; 50 Hz; 1ph ;3W, AC Supply (From UPS system)	
2.4.16	Space heater supply	:	230V, 50HZ, 1ph, 3W, AC Supply	

2.5 LOAD CLASSIFICATION

Electrical loads are classified in following categories based on process load criticality.

The following classification is based on allowable power disturbances or interruption for continuous operation of plant.

- Critical Load C1
- Semi Critical C2
- Non Critical Load C3

2.5.1 Critical load "C1"

Critical load "C1" is load which if interrupted (even momentarily) will cause an upset in the continuity of the process. Critical loads shall be fed through Un-interrupted power supply system.

2.5.2. Semi-Critical Load "C2"

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Semi-Critical load can be interrupted for small duration of 2 to 3 minutes without causing any upsets to the process. Semi-critical loads shall be fed through standby Diesel generator set upon normal / grid supply failure.

2.5.3. Non-Critical Load "C3"

Non-Critical load can be interrupted without causing any upsets to the process.

2.6 HAZARDOUS AREA CLASSIFICATION

Hazardous Area is considered to be an area where an explosive atmosphere is or maybe expected to be present which requires special precautions to be taken for the construction, installation and use of electrical equipment.

Hazardous Areas are classified into three Zones in which an explosive atmosphere may exist. Zones are defined as follows :

- Zone 0 - An area where an explosive atmosphere is present continuously.
- Zone 1 - An area where an explosive atmosphere is likely to be present at any time under normal operation.
- Zone 2 - An area where an explosive atmosphere is likely to be present only under abnormal operating conditions and for short periods

The areas outside those Classified as hazardous are referred to as "safe areas".

2.6.1 The Electrical equipment located in Hazardous areas shall be as follows :

HAZARDOUS AREA ZONE 0

- a) ~~(Ex)~~ ia - Intrinsically Safe
- b) ~~(Ex)~~ s - Special protection

HAZARDOUS AREA ZONE 1

- a) Equipment suitable for Zone 0
- b) (Ex) d - Flame Proof
- c) (Ex) e – Increased safety

HAZARDOUS AREA ZONE 2

- a) Equipment suitable for Zone 0 or Zone 1
- b) (Ex) n – Non sparking type (Provides protection against auto-ignition of surrounding gases / vapors).

2.7 The following areas are considered as Hazardous

- a) Speciality Building : Zone-1, Gas group C, Temp class T3
- b) Elastomer Building : Zone-1, Gas group C, Temp class T3
- c) Ware house (Partly) : Zone-0, Gas group X, Temp class T4
- d) Raw material storage tank area : Zone-2, Gas group D, Temp. class T4
- e) Fuel storage yard : Zone-2, Gas group D, Temp. class T4

Note : To be confirmed after Hazop study

3.0 DESIGN PHILOSOPHY

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3.1 POWER DISTRIBUTION PHILOSOPHY

3.1.1 PRIMARY DISTRIBUTION

Power supply to the plant shall be availed from Tamil Nadu Electricity Board (TNEB) Oragadam substation located approx. 3Kms from the proposed plant site through 11kV overhead transmission line.

The 11kV over head transmission line is received at 11kV Outdoor switchyard at proposed pashmina project site.

Outdoor Switchyard shall consist of : -

- a) One set of Lightning arrestors, Disconnecting switches, Current transformer, Potential transformer for TNEB protection & metering.
- b) 11KV outdoor kiosk with Off load break switch for local isolation by Consumer.

(Note : 11KV Breaker feeder at TNEB Oragadam substation, Transmission line, Switchyard equipment at plant premises, shall be sized considering loads of Phase I, & Technology center loads)

In addition to the grid supply the plant shall have Diesel generator sets of adequate capacity to cater power supply to Critical loads.

During normal operation power supply to the entire plant load shall be availed from TNEB. Upon failure of TNEB / Grid supply the Diesel generator set shall supply power to all Critical loads.

11KV Grid supply received at Outdoor switchyard shall be fed to 11KV Load break switch for local isolation & shall feed power to 11KV Switchboard-1 located at substation.

11KV Switchboard-1 shall have one incomer fed from Grid supply with required no. of outgoing for present loads & Technology center load. 11KV Switch board-1 shall feed power to distribution transformer for further distribution.

3.1.2 SECONDARY DISTRIBUTION

Two nos. of Power and motor control center (PMCC) is envisaged for secondary power distribution. Power supply to each PMCC shall be fed from two distribution transformers to cater present load & one incomer from DG supply

Each PMCC shall be fed from two 11/0.433KV Distribution transformer. In normal condition both the transformers of each PMCC shall be in working condition & each transformer shall be loaded to 50% of transformer capacity & upon failure / maintenance of one transformer, the other transformer shall be loaded 100% of the transformer capacity.

Each PMCC shall have Three Incomers & two bus couplers. PMCC shall feed power to all normal & Critical loads. Critical and Normal loads shall be fed from their respective PMCC / MCC.

Upon failure of grid supply, power is made available to all Switchboards irrespective of the type of loads (Critical, Normal). Depending on the market /process requirement the required loads can be switched on manually within the DG capacity. However DG shall be sized for Critical loads only. REV.

Upon resumption of Grid supply, DG supply shall be momentarily paralleled with Grid to avoid interruption of power supply during transfer of power supply from DG to Grid supply.

3.2 CABLES

HV Power cables shall be 11/11KV (UE) Grade, Multi-stranded copper conductor, XLPE insulated with extruded semi-conducting compound, with conductor & insulation screening, armoured, PVC outer sheathed, Halogen free, Flame retardant cables

LV Power & control cables shall be 1100V Grade, Multi-stranded Copper/Aluminium conductor, steel wire / strip armoured "OR" unarmoured, Halogen free, Fire retardant, cables.

Sl. No.	Cable & size	Conductor material	Remarks
1	Upto 16 Sq. mm.	Copper	
2	25 Sq. mm & above	Aluminium	

Cables shall be selected on the basis of current carrying capacity, permissible voltage drop, derating factors etc.

3.2.1 VOLTAGE DROPS

The maximum steady state voltage drops in various sections of the electrical system shall be within the limits as stated below :

a)	Outdoor switchyard to MV switchboard	0.5%
b)	MV switchboard to Distribution transformer	0.5%
c)	Cables between Distribution transformer to PMCC (Close by)	0.5%
d)	Cables between PCC and MCC (Close by) (Voltage drop upto motor shall be limited to 5%)	2%
e)	Cables between PMCC / MCC (situated near PCC) and motors	5%
f)	Cables between MLDB and lighting panel	1-1.5%
g)	Circuit between lighting panel and lighting points	4%

The maximum voltage drops during motor start-up / acceleration condition shall be within the limits stated below :

SYSTEM ELEMENT	OPERATING CONDITION	MAX. PERM.
----------------	---------------------	------------

		VOLTAGE DROP.	REV.
At the busbars of the worst affected switchboard / MCC	Start-up of large motor with other loads on the bus or acceleration of a group of motors	10%	
Cables from MCC to motor	Motor start-up or reacceleration	15%	

3.2.2 CABLING SYSTEM

- | | | |
|----|----------------------|--|
| a) | Process area | Overhead along pipe racks / Lined concrete trench / Embedded electrical grade pipes / pipe sleepers. |
| b) | Overall plant area | Overhead along pipe racks / Duct banks / Lined concrete trench / Directly buried / Pipe sleepers |
| c) | Offsite paved area | Overhead cable tray / Lined trench |
| d) | Offsite unpaved area | Overhead cable tray / Lined trench / Directly buried |
| e) | Type of cable trays | Galvanized prefabricated.- Ladder type |
| f) | Cable tray supports | GI Prefabricated "OR" Site fabricated. |

3.2.3 CABLE GLANDING & TERMINATION

- | | | | |
|----|--|---|---------------------------|
| a) | Indoor Panel | : | Single compression |
| b) | Motors in field | : | Double compression |
| c) | Push buttons and control Panels, | : | Weather / explosion proof |
| | Light fittings, JB's LDB's etc.in Field. | : | Double compression |
| d) | Type of Cable Lugs (Tinned Cu / Al.): | | Tinned Copper / Al. |

3.3 ELECTRICAL ROOMS / TRANSFORMER AREA

3.3.1 ELECTRICAL ROOMS – GENERAL REQUIREMENTS

- | | | | |
|----|---|---|-----------------|
| a) | All electrical rooms shall be clean, dry, well sealed and free of dust, vapor, and vermin to provide reliable operation of equipment. | | |
| b) | Each electrical room shall be provided with two doors. | | |
| c) | Each electrical room shall be equipped with fire alarm system & fire fighting facilities. | | |
| d) | <u>ENVIRONMENTAL CONDITIONS</u> | | |
| | Inverter room / Power UPS Room | : | Air conditioned |
| | Electrical room | : | Pressurized |
| | Control Room | : | Air Conditioned |

3.3.2 TRANSFORMER AREA – GENERAL REQUIREMENTS

- | | | | |
|----|-----------------|---|-------------------------|
| a) | Height of fence | : | Minimum 6 Ft. from FFL. |
| b) | Clearance | : | All around 1000 mm. |

3.4 GROUNDING AND LIGHTNING PROTECTION

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- a) Grounding system shall be according to IS 3043 and local rules & regulations.
- b) Protective grounding and service grounding shall be connected together to create common grounding system with common grounding electrode.
- c) Main DCS ground bus bar shall be connected to the DCS ground electrode by two separate PVC insulated copper conductors.
- d) Grounding rods & conductors shall be designed based on soil resistivity and maximum ground fault current. The ground conductor shall be adequately sized to withstand the maximum fault current for 1 sec without any deformation and with tolerable temperature rise.
- e) Steel reinforcing bars of concrete structure of the building (Columns, beams, foundation and footing) shall be connected all together by welding to achieve electrical continuity. If steel reinforcing bars do not have enough size for maximum ground fault current they shall be augmented by additional ground conductor.
- f) G.I. strip shall be installed in concrete of grade slab to connect all columns to create one common grounding electrode. Reinforcement bars of each column shall be connected to the grid of G.I. strip by welding.
- g) Grounding Electrode type
 - Transformer Neutral Grounding : Copper plate type Electrodes
 - Body grounding : GI pipe electrode
 - DCS grounding : Copper rod / DCS vendors recommendation.
- h) Main earth loop material : GI strip
- i) Ground Resistance : 1 to 2 Ohms
- j) Substation earth loop : GI / copper
- k) Lightning conductor : Copper

3.5 START / STOP PHILOSOPHY

- 3.5.1 All non-package motors shall have Local push button station, consisting of start push button, emergency stop push button & local remote selector switch as per operational/process requirement. Manual – Off – Auto selector switch – lockable type shall be provided for Local & remote operation of motors.

Generally during plant operation Manual – Off – Auto selector switch position shall be in AUTO mode. During Maintenance / testing selector switch position shall be changed to MANUAL mode & shall be able to start / stop the motor by pressing start / stop push button at Local control station.

- 3.5.2 Stop push button facility shall be provided in MCC for each motor.

- 3.5.3 Packaged equipment drives which require automatic control from the Vendor control panel shall be controlled locally.

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3.6 CONVENIENCE RECEPTACLES

3.6.1 415V, 63A, 5Pin, TP&N+E shall be provided considering accessibility with 50 Meters cable length.

3.6.2 230V, 16A, 3pin SP&N+E shall be provided considering accessibility with 25 Meters cable length.

4.0 ELECTRICAL EQUIPMENT**4.1 11 KV OUTDOOR SWITCHYARD EQUIPMENTS****4.1.1 11 KV OUTDOOR LOAD BREAK SWITCH**

a) Type	:	Off load
b) Rated voltage	:	11KV ± 10 %
c) Rated frequency	:	50 HZ ± 3 %
d) Rated current	:	400A
e) Short circuit level at 11kV	:	13.1 KA (To be confirmed later)

4.1 EMERGENCY (STAND BY) DG SET**4.2.1 GENERAL**

a) Number of Generating Sets	:	One OR Two
b) Service	:	Standby
c) Voltage	:	415V
d) Frequency	:	50 HZ ± 3 %
e) Fault Level	:	Later
f) Change over Scheme (AMF /Manual):	:	Auto mains failure
g) Time within which power must be made available to essential loads	:	Immediate
h) Sequences of motor starting	:	Not required
Mode of operation	:	
- Independent	:	Yes
- Synchronised	:	Momentary paralleling with Grid
k) Neutral Grounding	:	Solidly Grounded
l) Accoustic enclosure	:	Required.
m) Export of Power to Grid	:	Not Envisaged
n) Future provision for Export of power to Grid	:	Not Required

4.2.2 ENGINE

a) Speed of the engine	:	Later
b) Type of starting (Battery / Air)	:	Battery
c) Type of Cooling of the Engine	:	Radiator cooled
d) Type of Fuel Oil	:	HSD
e) Storage of Fuel Oil	:	990Litres capacity
f) Requirement of Utilities	:	

- | | | | |
|---|---|----------|------|
| - Cooling Water | : | Required | REV. |
| g) Maximum Rating of the Drive to be started on the DG set | | | |
| - Rating KW | : | Later | |
| - Type of starting | : | Later | |
| h) Whether the Drive is started with | : | Both | |
| No Load or on Load | | | |
| i) Time within which DG can be fully loaded from the instant of starting signal given | : | 90 sec | |

4.2.3 ALTERNATOR

- | | | |
|---------------------------------------|---|--|
| a) Alternator capacity | : | Present process Critical loads only |
| b) Type of Excitation | : | Brushless |
| c) Suitability for parallel Operation | : | Required |
| d) Type of Cooling | : | Air Cooled |
| e) Type of enclosure | : | IP-23 |
| f) Temperature Detecting elements | | |
| - For Winding | : | Later |
| - For Bearing | : | Later |
| g) Number of Bearings | : | As per Vendor's recommendation |
| h) Class of insulation | : | Class – F with temp. rise limited to class B |

4.3 DISTRIBUTION TRANSFORMER

- | | | |
|---|---|---|
| a) Voltage Ratio at No Load | : | 11./ 0.433kV |
| b) Spare Capacity considered | : | Operating load not beyond 80% |
| c) Tap Changer | : | Off Circuit |
| d) Range of Tap Changing | : | ± 5% at 2.5 % per step. |
| e) Cooling | : | ONAN |
| f) Termination | | |
| - Primary Termination | : | Cable |
| - Secondary termination | : | Bus duct |
| g) Neutral Ground Arrangement | : | Solid grounding |
| h) Vector Group. | : | Dyn 11 |
| i) Special Accessories such as MOG, WTI, OTI, Gas relay | : | Required |
| j) Temperature rise of oil / winding | : | 50/55°C (by oil / winding resp.) over ambient temp. |
| k) Winding Material (Cu / Al) | : | Copper |
| l) Impedance with tolerance | : | As per IS |
| m) Painting | : | RAL 7032 |

4.4 LV BUSDUCT

- | | | |
|---------------------------------|---|--------------------------------|
| a) Type & section | : | Non-segregated – Air Insulated |
| b) Bus duct section | : | Rectangular |
| c) Degree of protection | : | Indoor : IP 54, Outdoor : IP55 |
| d) Continuous rating of Busbars | : | Later |
| e) Fault level | : | Later |

f) Voltage grade	:	415 Volts	REV.
g) Busbar material	:	RAL 7032	
- Phase & Neutral	:	Aluminium	
- Earth Busbar	:	Aluminium "OR" GI	
h) Copper flexible connections	:	PMCC end & Transformer end	

4.5 POWER & MOTOR CONTROL CENTER / MOTOR CONTROL CENTER

4.5.1 Construction :

a) Intelligent / Non intelligent	:	Non-Intelligent
b) Single \ double front	:	Single front
c) Execution Drawout / fixed	:	Drawout
d) Degree of protection	:	IP 21
e) Spare feeders	:	10%
f) Paint Shade	:	RAL 7032

4.5.2 Type of Circuit Breaker

a) Drawout / Non – Drawout	:	Drawout,
b) TPN / 4 Pole	:	TPN
c) Direct Acting Releases/with relays:	:	μP based releases.

4.5.3 Busbar

a) Material	:	Aluminium
b) Insulation of Busbars	:	PVC sleeved
c) Fault Level	:	Later.
d) Temperature Rise above Max. Ambient of 47°C	:	50°C (final temp. in fault condition should not go beyond 97°C)

4.5.4 Incoming (Cable / Busduct)

a) PMCC	:	LV Busduct
b) MCC.	:	Cables

4.5.5 Cable entry :

Bottom

4.5.6 Isolator type (MCCB / SFU / ACB)

a) Upto 630A	:	MCCB
b) 800A and above	:	ACB

4.5.7 Type of Motor starters

a) DOL starting	:	Upto & including 55 KW
b) Star delta starting	:	75 KW and above

4.5.8 Protective Relays

a) Incoming Feeders	:	μP based release
b) Bus Coupler	:	μP based release
c) Feeder to MCC, MLDB, Capacitor panel	:	MCCB with built in release / Fuse

4.5.9 Metering

Incoming Feeder & outgoing feeders :	:	Digital Multi function meter
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4.5.10 Provision of Alarm / Tripping at DCS : Hard wired

4.5.11 Interfacing to DCS : Hard wired

4.6 MAIN LIGHTING DISTRIBUTION BOARD / POWER DISTRIBUTION BOARD

4.6.1 Construction

a) Single \ double front	:	Single front
b) Execution Drawout / fixed	:	Fixed
c) Degree of protection	:	IP 21
d) Spare feeders	:	10% "OR" one of each rating
e) Paint Shade	:	RAL 7032
f) Compartmentalised / Non compartmentalized	:	Compartmentalized.
f) Material of Construction	:	CRCA sheet steel.
g) Floor / wall mounting	:	Floor
h) Cable Entry (Top / Bottom)	:	Bottom

4.6.2 Busbar

a) Material	:	Aluminium
b) Fault level	:	Later
c) Location	:	Top

4.6.3 Incoming / Outgoing Feeders

a) Type of Component	:	MCCB
b) Design (TPN / 4 Pole)	:	TPN
c) Metering (Voltage, Current)	:	Later

4.6.4 Location

: Indoor

4.7 INVERTERS

a) Input supply	:	415V, 3phase, 4Wire, AC Supply
b) Bypass arrangement	:	Direct-on-line / Star delta - As reqd.
c) Field current measurement	:	CT's to be provided
d) Degree of protection	:	IP 21
e) Rectifier	:	Diode Bridge
f) Inverter	:	IGBT with adjustable v/f profiles
g) EMI / RFI Suppression	:	Required
h) Line chokes	:	Required
i) Speed Control with Current & Torque limit:	:	Vector control software configured for open loop
j) Key pad for programming & operator panel:	:	Required
k) Analogue signals 4-20 mA	:	Speed reference, Speed control, Current Output, KW Output
l) Digital signals	:	DCS start / Stop, Run status, Fault alarm, Ready to start.
m) Power loss ride through facility	:	Required
n) Drives are suitable for long cable length	:	Approx. 300Mtrs.

o) Mounting	:	Floor/Wall "OR" housed in PMCC/MCC	REV.
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4.8 UN-INTERRUPTED POWER SUPPLY (UPS) SYSTEM

a) AC Input supply	:	415V ± 10 %, 50HZ, 3ph, 4Wire AC.
b) Output voltage & variation	:	110 V ± 2 %- For Instrument UPS
	:	415 V ± 2 %- For power UPS
c) Output frequency & variation	:	50 HZ ± 1 %
d) Rated KVA		
Instrument UPS	:	Later
Power UPS	:	Later (Phase I Critical loads.)
e) Static Transfer switch	:	Required
f) Audio visual alarm	:	Required
g) AC distribution board	:	Required
h) Battery type	:	Lead acid Sealed low maintenance
i) Duration of back-up for rated output	:	
- Power UPS	:	6 Minutes
- Instrument UPS	:	30 Minutes

4.9 BATTERY, BATTERY CHARGER

a) AC Input supply	:	415V ± 10 %, 50HZ, 3ph, 4Wire AC.
b) AC input frequency	:	50 HZ ± 3 %
c) DC output voltage settings		
- Nominal	:	110V
- Float	:	121V
- Boost	:	148V
d) Power conversion	:	AC to DC by means of three phase thyristor controlled rectifier
h) Battery type	:	Nickel-Cadmium
i) Future load growth	:	For phase II loads
i) Duration of back-up for rated output	:	8 Hrs.

4.10 MV / LV MOTORS

a) Type of Construction (Squirrel Cage / Slip ring)	:	Squirrel Cage
b) Type of Cooling	:	TEFC
c) Design (Standard / "d" / "e" / n)	:	As required
d) Mounting (Horizontal / Vertical)	:	As required
e) Temperature Detector in Winding	:	For MV Motors (If any)
f) Temperature Detector in Bearing	:	For MV Motors (If any)
g) Space Heater in Motor	:	75 KW and above
h) Degree of protection	:	IP 55
i) Type of starting	:	As Applicable
j) Method of Speed Variation, if any	:	Inverter
k) Type of Coupling	:	Different for different equipment
l) Insulation	:	Class – F with temperature rise limited to Class – B

m) Painting	:	RAL-7032	REV.
o) Service Factor			
• Upto 185 kW (Fixed speed motors)	:	1.15	
• Inverter driven motors	:	1.0	
p) Type of Starters for Motors			
• DOL starting	:	Upto & including 55 KW	
• Star delta starting	:	75 KW and above.	
q) Terminal Box			
• Power cables	:	Required	
• Space heaters	:	For motors 75KW & above.	

4.11 LOCAL CONTROL STATION

a) Material of Construction		
Weather proof	:	CRCA sheet steel/FRP.
Flame proof	:	Cast Aluminium LM6
b) Degree of protection		
Weather proof	:	IP 55 - Outdoor Installation.
	:	IP 21- Indoor Installation.
Flame proof	:	IP 55 (Minimum)
c) Colour shade	:	RAL-7032
d) Size of Indicating Meters	:	Later
e) Canopy	:	Required for Outdoor installation
f) Ammeter	:	Motors rated 15KW & above
g) Selector switches	:	As per control scheme

4.12 FLAME PROOF / EXPLOSION PROOF EQUIPMENT :

Flame proof / Explosion proof equipment shall be installed in Hazardous location. All flame proof equipment shall be of type already tested from CMRI and approved by CCE / DGMS / DGFASLI (as required) for use in specified gas group locations.

4.13 PACKAGE UNITS EQUIPMENT

- a) Package unit equipment shall follow requirements from this Electrical Design Basis.
- b) Power Supply will be provided at one point control supply shall be derived from the same source. Control panels to be supplied by the vendor as a part of the package.
- c) Generally all the engineering for package unit viz. Cable tray layout, Cable MTO, Tray MTO, Miscellaneous MTO, etc. shall be carried out by the package vendor. However, in the event that the same is to be carried out by Chemtex, the package vendor shall furnish motor and heater list, equipment layout and the cable list in the format mutually agreed with Chemtex.

5.0 LIGHTING SYSTEM

5.1 LIGHTING DESIGN PHILOSOPHY

REV.

- a) Lighting design shall conform to relevant IS Codes, International Codes and Standards, IES Hand Book and shall take into consideration, the requirements of safety and ease in operation and maintenance. A maintenance factor of 0.7 shall be assumed for lighting level calculations for normal areas. However, for dusty areas, maintenance factor as per relevant codes and standards shall be followed.
- b) Power supply to Sub-lighting distribution board is fed thru PMCC, upon grid supply failure DG set shall supply to all Lighting distribuion boards within 2 to 3Minutes.
- c) 20% of normal Lighting fixtures fed from Sub-lighting distribution boards (fed thru PMCC) shall have battery backup of 30Minutes & shall be switched on immediately during power failure.
- d) Each Lighting circuit from Sub-Lighting distribution board shall be loaded upto 1000 Watts / circuit.
- e) Location of Light fittings with battery Back-up : Required at strategic locations like stairway, Substation, Entrance etc.
- f) Lighting Junction boxes type
- Indoor – safe area : GI conduit / WP cast iron)
 - Indoor – Hazardous area : Explosion proof,
 - Outdoor area- : WP Cast iron / Explosion proof
- g) CONTROL PHILOSOPHY
- Street lighting / Pipe rack area (Auto / Manual) : Auto
 - Outdoor process area (Auto / Manual) : Auto
 - Process building area (Auto / Manual) : Auto
 - Enclosed Process building area (Auto / Manual) : Manual
 - Outdoor Switchyard : Auto
 - Electrical substation : Manual
 - Control Room : Manual
 - Office, Lab, Security, Ware house : Manual

Areas with Auto control shall be controlled through Synchronous timer / Photo cell.

- h) Power supply to Street / Area lighting shall be thru solar cells.
- i) ILLUMINATION LEVELS & TYPE OF LAMPS.

Given below are the recommended illumination levels and type of fittings for different area with its classification.:

Area	Area Classification	Lux Level	Type of Lamps
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Roads	Non Hazardous	20	HPSV	REV.
Tank Farm	Hazardous	100	MLL / HPMV	
Boiler Building	Non hazardous	100	HPMV	
Utility Building	Non hazardous	150	HPMV	
Fire water pump house	Non hazardous	150	HPMV	
Elastomer Building, Speciality Building	Hazardous	150	HPMV	
Outdoor Process Area	Non Hazardous	20	HPMV / HPSV	
	Hazardous (*)		HPMV / HPSV	
Offices, Security block.	Non Hazardous	300	Fluorescent	
Laboratory & QC	Non Hazardous	300	Fluorescent	
	Hazardous (*)		Fluorescent / HPMV	
Canteen / Change Room	Non Hazardous	150	Fluorescent	
Rest rooms, Security block	Non Hazardous	200	CFL / Fluorescent	
Work shop / Maintenance Room	Non Hazardous	150	Fluorescent / HPMV	
Control Room	Non Hazardous	300	Fluorescent	
Substation	Non Hazardous	200	Fluorescent	
Substation Cable cellar	Non Hazardous	100	Fluorescent	
Warehouse & Storage	Non Hazardous	150	Fluorescent / HPMV	
	Hazardous (*)		Fluorescent / HPMV	

Note :

* - Indicates "Will be confirmed after HAZOP study"

5.2 WIRING TYPE

- a). Process plant / outdoor area : Armoured / Unarmoured cable
- b). Buildings with false ceiling : Unarmoured cable / GI conduit
- c). Substation : Unarmoured cable / GI conduit / channel
- d). Grounding of Light Fitting : 3rd Core of Cable
- e). Type of Cable (Cu / Al) : Copper (Minimum) 2.5mm²
- f). Voltage Grade of Wire / Cable: 660V grade

5.3 SUB LIGHTING DISTRIBUTION BOARD

5.3.1 Construction

- a) Single \ double front : Single front
- b) Execution Drawout / fixed : Fixed
- c) Degree of protection

- Weather proof : IP 55 - Outdoor Installation.
- : IP 21- Indoor Installation.
- Flame proof : IP 55 (Minimum)

	d) Spare feeders	:	10% "OR" one of each rating	REV.
	e) Paint Shade	:	RAL 7032	
	f) Compartmentalised / Non compartmentalized	:	Non compartmentalized.	
	g) Material of Construction	:		
	-Weather proof	:	CRCA sheet steel.	
	-Flame proof	:	Cast Aluminium LM6	
	h) Floor / wall mounting	:	wall	
	i) Cable Entry	:	Top / Bottom	
5.3.2	Busbar			
	a) Material	:	Copper	
	b) Fault level	:	10kA for 1 Sec.	
5.3.3	Incoming Feeder			
	a) Type (TPN /DP/SP)	:	TPN	
	b) Type of Component	:	ELCB with 300mA sensitivity	
5.3.4	Outgoing Feeder			
	a) Type (SPN /DP)	:	SPN , MCB	
	b) Component	:	MCB	
6.0	<u>HEAT TRACING SYSTEM</u>	:	Not Envisaged	
7.0	<u>FIRE ALARM SYSTEM</u>			
7.1	Fire alarm panel			
	a) Location	:	Main panel at Control Room & Repeater panel at Security Main gate	
	b) Mounting type	:	Wall	
	c) Standby power	:	24V DC sealed maintenance batteries	
	d) Comaptability	:	Smoke detectors, Heat detectors, Manual call points	
7.2	Detector type	:	Addressable analogue	
7.3	Power supply	:	230V, 50 HZ, AC Supply	
7.4	Hooter type & range	:	Electronic, 85 Db	
7.5	No. of Zones	:	12 expandable upto 24 zones	
8.0	<u>PLANT COMMUNICATION SYSTEM</u>			
8.1	<u>PUBLIC ADDRESS SYSTEM</u>			
	a) PA system type	:	Modular design, microprocessor based	
	b) No. of Zones	:	10 Expandable upto 20	
	c) Interfacing	:	Telephone system	
	d) Power supply	:	110V, 50HZ, AC Supply	
	e) Loud speaker type	:	Horn type – Safe & hazardous area	
		:	Recessed type – Suitable for false ceiling (If any)	
	f) Loud speaker range	:	30Mtrs	
	g) Location	:	Control room	
	h) Areas for paging	:	Speciality, Elastomer, Boiler, Utility, Ware House, Substation & Control Room.	

8.2 TELEPHONE SYSTEM**REV.**

a)	EPABX type	:	Modular design, microprocessor based.
b)	No. of P & T lines	:	24 Expandable upto 32 lines
c)	No. of extensions	:	50 Extensions, expandable upto 75
d)	Interfacing	:	Industrial paging system
e)	IP Phones	:	6 (six) *
f)	Power supply	:	110V, 50HZ, AC Supply
f)	Battery backup	:	30Mins.
g)	Battery type	:	Maintenance free
h)	Telephone types	:	Basic telephone, Speaker telephone, Caller ID telephone, Explosion proof
i)	Cable type	:	Cat -6, Jelly filled armoured / unarmoured cable
j)	Location	:	Administration building

Note :

1. P & T lines & EPABX is considered only for the proposed Pashmina project, separate provision is to be considered for proposed Technology center.
2. Exact quantity of IP phones shall be decided during purchase order.

9.0 STATUTORY APPROVAL :

Following statutory clearance, liason, approval from Government agencies shall be by Momenitive : -

- a) Tamil Nadu Electricity Board (TNEB) to obtain statutory approval to avail power supply
- b) Pollution control Board for DG Installation
- c) Chief controller of explosives (CCOE) statutory authority for type of Ex equipments used in Hazardous areas.
- d) Telephone Department to avail telephone lines
- e) Department of Communication to avail permission for Public address system.